

Table of Contents

1	Motivation and Objectives	3
2	System Design	3
2.1	Key Innovations	3
2.2	Comparison with Existing Solutions	4
2.3	Potential Applications	5
3	Methodology and Implementation	5
3.1	System Overview	5
3.2	Cardiovascular Risk Prediction	7
3.2.1	Dataset and Problem Setup	7
3.2.2	Preprocessing and Filters	9
3.2.3	Feature Extraction Methods	9
3.2.3.1	Maximum Heart Rate	9
3.2.3.2	Oldpeak	9
3.2.3.3	ST Slope	10
3.2.3.4	Resting Electrocardiogram (RestingECG)	10
3.2.3.5	Validation:	11
3.2.4	Model Training Methodology and Techniques	12
3.2.4.1	Feature Preprocessing	12
3.2.4.2	Training	12
3.2.4.3	Classification	13
3.3	AF Detection Methodology and Validation	13
3.3.1	RdR Map Construction	14
3.3.2	Gridding and Non-empty Cells (NEC)	14
3.3.3	Threshold Determination	14
3.3.4	Edge Deployment Advantages	14
3.4	TAIDE-based health assistant	15
3.5	System Deployment, Cost Analysis, and Feasibility	16
3.6	Prototype Implementation and System Demonstration	17
4	Research Outcomes and Benefits	20
4.1	Enhancement of Early Detection Capability of Cardiovascular Abnormalities	20

4.2 From Raw ECG Signals to Interpretable Risk Representation	21
4.3 Lightweight Edge Backend Deployment	22
5 Conclusion and Future Work	23
5.1 Conclusion	23
5.2 Future work	24
6 References	24

1 Motivation and Objectives

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, highlighting the critical need for effective continuous monitoring and early detection. Recent advances in wearable technologies have made electrocardiogram (ECG) monitoring increasingly accessible in daily life.

However, **most existing wearable ECG systems are limited to short-duration recordings**, typically around one minute, and provide only basic real-time analysis, such as heart rate estimation. While some devices support arrhythmia detection, their functionality is often primarily limited to detecting persistent atrial fibrillation (AF), without the capability to identify more complex or transient conditions.

Atrial fibrillation (AF) is commonly classified into paroxysmal, persistent, and permanent forms according to episode duration and termination behavior [1]. Paroxysmal AF refers to episodes that terminate spontaneously, typically within several days, whereas persistent and permanent AF require clinical intervention or cannot be terminated. Due to its intermittent nature, **paroxysmal AF is particularly challenging to detect using conventional short-duration monitoring systems**, and such episodes are therefore frequently missed. This limitation is critical, as early detection of transient arrhythmias and comprehensive cardiovascular risk assessment are essential for timely medical intervention and long-term disease prevention.

Therefore, we propose an **integrated framework that enables long-term ECG monitoring** and incorporates two components: an **atrial fibrillation detection module** and a **cardiovascular disease prediction model**. The AF detection module focuses on identifying both paroxysmal and persistent atrial fibrillation from continuous ECG signals, while the cardiovascular disease prediction model provides a comprehensive assessment of cardiovascular conditions based on extracted features. Our proposed system offers a more complete solution for both real-time arrhythmia detection and cardiovascular risk evaluation.

2 System Design

2.1 Key Innovations

The proposed system introduces a unified framework for continuous ECG monitoring and analysis, integrating long-term data acquisition, real-time signal processing, and edge-based inference. Its key features are summarized as follows:

- **Interpretable physiological indicators:** The system incorporates cardiovascular risk assessment derived from ECG features, transforming raw signals into meaningful and interpretable outputs accessible to non-specialist users. In addition, it supports the export of ECG signal plots to facilitate subsequent review by healthcare professionals.
- **Long-term continuous monitoring:** Supports sustained ECG acquisition beyond the short-duration measurements of typical wearable devices.
- **Comprehensive AF detection:** Capable of detecting both paroxysmal and persistent AF.
- **Multi-condition ECG acquisition:** Enables ECG monitoring under both resting and exercise conditions, supporting more realistic usage scenarios.

The system is implemented using an edge-based architecture to achieve real-time inference with low latency and enhanced data privacy.

2.2 Comparison with Existing Solutions

Our system enables **long-term, continuous ECG monitoring**, representing a substantial advancement over typical commercial wearable devices that are generally restricted to short-duration recordings.

In addition, **the system incorporates cardiovascular risk assessment**, thereby enhancing the interpretability and clinical relevance of the acquired ECG signals.

Regarding AF detection, the proposed system is **capable of identifying both paroxysmal and persistent AF**, whereas many commercial devices primarily focus on heart rate monitoring or are limited to detecting persistent AF.

Furthermore, **the system is specifically designed for wearable, continuous operation and supports ECG acquisition under both resting and exercise conditions**. In contrast, many existing commercial devices rely on finger-contact measurements and are therefore limited to short-term, resting-state recordings.

Table 1: Functional comparison with commercial products

Comparison Metrics	Proposed System	Existing Wearable Devices
ECG Monitoring Duration	Long-term continuous monitoring	Short-duration recordings
ECG Acquisition Conditions	Resting and exercise ECG	Primarily resting ECG
Measurement Method	Gel electrodes	Dry electrodes (finger-contact required)
Analytical Functionality	ECG feature analysis, risk assessment, and AF detection	Heart rate monitoring or limited AF detection
AF Detection Capability	Paroxysmal and persistent AF	Persistent AF detection

Computational Architecture	Edge-based real-time inference with enhanced privacy	Device-dependent; often limited on-device processing
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2.3 Potential Applications

For general users, the system provides interpretable insights into cardiovascular health status and potential risk factors, enabling timely preventive actions and early intervention. For healthcare professionals, it functions as an effective remote monitoring tool, supporting continuous observation of ECG signals and facilitating the prompt identification of clinically significant abnormalities or temporal changes.

Moreover, owing to its low-cost implementation and edge-based architecture, the system can be readily deployed across diverse settings, including community healthcare facilities and resource-constrained environments, thereby improving the accessibility and scalability of cardiovascular disease screening and management.

3 Methodology and Implementation

3.1 System Overview

The system consists of the following components: AD8232 ECG module, ESP32, and Raspberry Pi 3B. The AD8232 is responsible for single-lead ECG signal acquisition, while the ESP32 connects for data sampling and transmission, and the Raspberry Pi 3B receives those data for signal processing, feature extraction, AF detection, risk inference, and result integration. **This architecture can meet the cost, portability, and edge deployment requirements,** making it suitable for use in remote health monitoring applications where efficient and real-time ECG analysis is essential. Figure 1 shows how these components interact within the overall system architecture and the services inside each module.

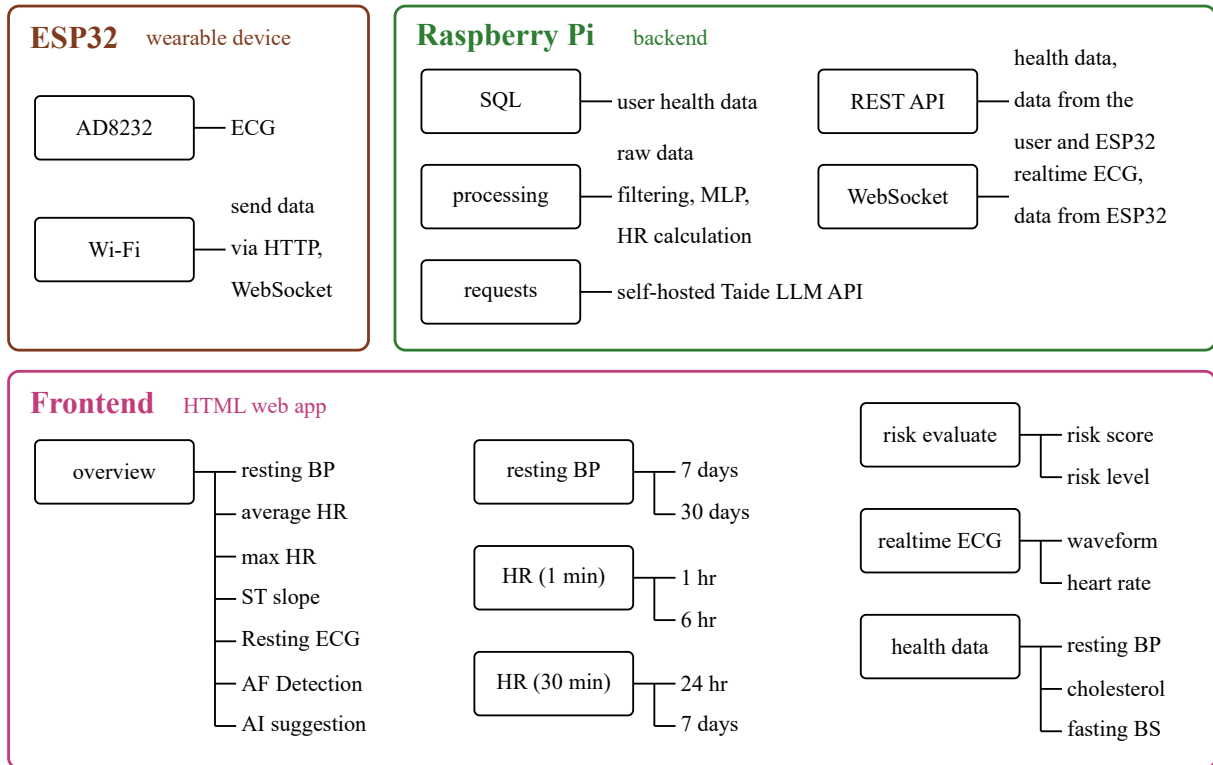


Figure 1: Module overview

- **Preprocessing:**

The raw ECG signal is first preprocessed to enhance signal quality. This stage includes noise reduction, baseline wander removal to ensure robustness in subsequent analysis.

- **R-peak Detection:**

R-peaks are detected using the Pan-Tompkins++ algorithm [2], providing accurate localization of cardiac cycles. These detected peaks serve as the basis for both feature extraction and rhythm analysis, as shown in Figure 2.

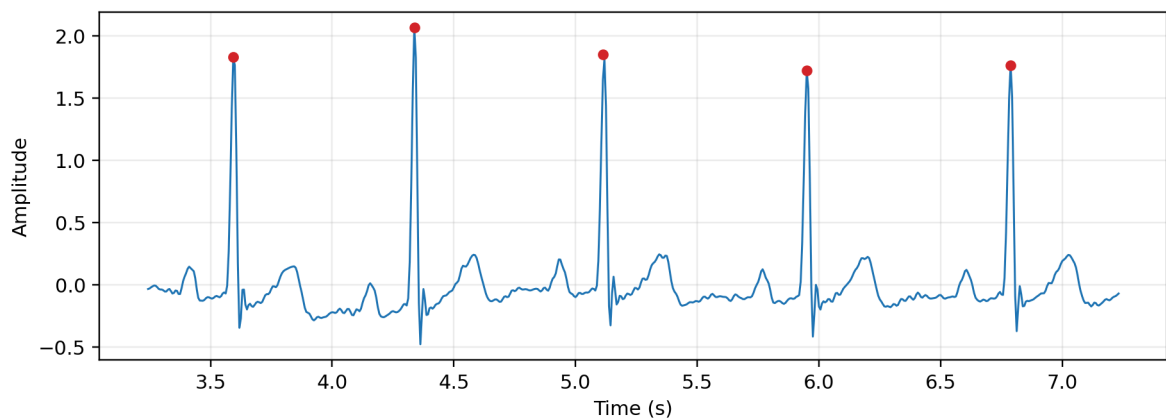


Figure 2: Illustration of R-peak detection on the ECG signal

- **Feature Extraction:**

Based on the detected R-peaks, features are extracted, including maximum heart rate

(MaxHR), Oldpeak, ST_Slope, and Resting ECG indicators (RestingECG). These features are used for downstream cardiovascular risk evaluation.

- Parallel Analysis Modules:

After R-peak detection, the pipeline branches into two parallel modules:

- Risk Assessment Model:

The extracted features are input into a CVD risk assessment model to generate a risk score representing the likelihood of underlying cardiac conditions.

- AF Detection:

Based on the RdR+NEC algorithm [3], our system constructs an RdR map by plotting consecutive RR intervals versus their beat-to-beat differences (dRR) to capture rhythm irregularity. Classifying AF by counting the number of non-empty cells (NEC) on a 25-ms resolution 2D grid over a 128-beat window.

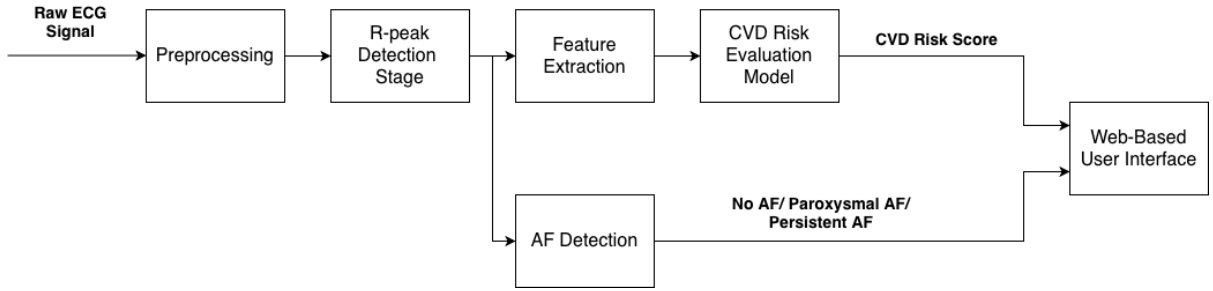


Figure 3: Data flow of the system

3.2 Cardiovascular Risk Prediction

3.2.1 Dataset and Problem Setup

We utilized cardiovascular disease datasets from the UCI repository, including the Cleveland, Hungary, Switzerland, and VA Long Beach databases, and further integrated them with the Stalog (Heart) dataset. After removing duplicate entries and records with missing values, the final dataset comprised 918 samples. It includes 11 input features and one target variable, as detailed in the following table.

Table 2: Summary of Features for Heart Disease Dataset

Feature	Description
Age	Age of the patient (years)
Sex	Sex of the patient (M: Male, F: Female)
ChestPainType	Chest pain type (TA: Typical Angina, ATA: Atypical Angina, NAP: Non-Anginal Pain, ASY: Asymptomatic)

RestingBP	Resting blood pressure (mm Hg)
Cholesterol	Serum cholesterol (mm/dl)
FastingBS	Fasting blood sugar (1: >120 mg/dl, 0: otherwise)
RestingECG	Resting electrocardiogram results (Normal, ST: ST-T abnormality, LVH: left ventricular hypertrophy)
MaxHR	Maximum heart rate achieved (60–202)
ExerciseAngina	Exercise-induced angina (Y: Yes, N: No)
Oldpeak	ST depression value (numeric)
ST_Slope	Slope of peak exercise ST segment (Up, Flat, Down)
HeartDisease	Output class (1: Heart disease, 0: Normal)

We conducted a feature ablation analysis to evaluate the contribution of each feature to the model’s predictive performance. Specifically, we compared the model accuracy when each feature was included versus excluded (See Figure 4). Therefore, we excluded this feature from the final deployed model (retaining 10 out of 11 features) to enhance robustness and generalization performance. Additionally, since `Cholesterol` and `FastingBS` cannot be directly measured by our device and require user input, we designed a **dual-model system**. The 10-feature model is applied when all features are available, whereas an **8-feature model is used when `Cholesterol` and `FastingBS` are missing**. This design maintains strong predictive performance while ensuring reliable operation under incomplete data conditions.

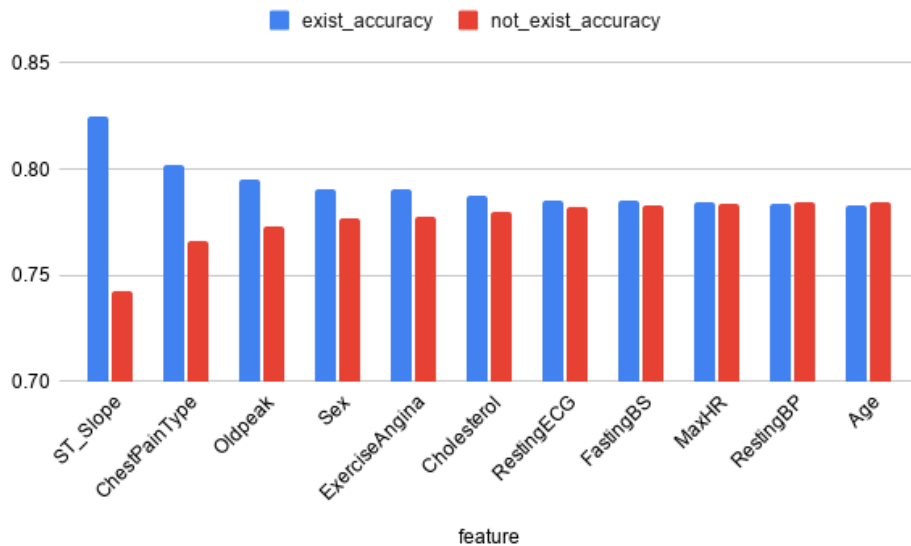


Figure 4: Ablation study on input features

3.2.2 Preprocessing and Filters

Before extracting features, raw ECG signals must undergo noise filtering to ensure the accuracy of subsequent feature calculations. Our project initially utilized a 5-12 Hz filter to process the QRS segment. However, this frequency band over-suppressed the T-wave, causing the false removal of genuine ST depression or ST elevation features. To address this, we designed an **independent 0.5-35 Hz filtering interval specifically for the ST Slope calculation**, preserving complete and genuine ST segment features.

3.2.3 Feature Extraction Methods

Our processing pipeline primarily extracts four key features: Maximum Heart Rate, Oldpeak, ST Slope, and Resting Electrocardiogram (RestingECG). The R-peak measured by Pan-Tompkins++ [2] serves as the reference point for calculating all subsequent features (like heart rate and segment windows). The extraction mechanisms, challenges, and optimization processes are detailed below.

3.2.3.1 Maximum Heart Rate

Definition: The maximum heart rate achieved during the measurement period, typically yielding a numeric value between 60 and 202.

Extraction Method: By utilizing the Pan-Tompkins++ algorithm, we were able to detect R-peaks, which are subsequently converted into the heart rate.

3.2.3.2 Oldpeak

Definition: Oldpeak represents the ST depression caused by activity in comparison to rest. A value > 0 may indicate angina, while a value < 0 is potentially related to myocardial infarction. Oldpeak quantifies the absolute displacement of the ST segment relative to the electrically neutral PR baseline. In a healthy heart, the ST segment should be perfectly flush with the PR segment. When the heart muscle is stressed or deprived of oxygen (especially during the exercise phase of stress testing), the ST segment will deviate.

Extraction Method: It is calculated based on relative R-peak positions using the following formula:

$$\text{Oldpeak} = V_{\text{ST level}} - V_{\text{PR baseline}}$$

Here, *PR baseline* is the average potential between 200 ms and 120 ms prior to the R-peak, and *ST level* is the average potential between 100 ms and 160 ms after the R-peak. This timing

window is crucial for accurately measuring the ST segment's displacement, and is suggested by multiple medical literature sources. [4], [5], [6]

3.2.3.3 ST Slope

Definition: The slope of the peak exercise ST segment, classified into three categories: Up, Flat, or Down. While the Oldpeak measures the absolute drop or rise of the ST segment, the ST_Slope measures its morphological trajectory. How the ST segment behaves dynamically over time is a critical predictor of coronary artery disease.

Algorithmic Execution: Instead of taking a single average, the algorithm extracts all voltage data points within the defined ST Window and applies an Ordinary Least Squares linear regression to find the line of best fit:

$$V(t) = mt + b$$

Here, t represents time, $V(t)$ is the voltage, b is the y-intercept, and m is the computed slope.

Diagnostic Classification: We categorize the ST slope into three categories: Up (upsloping), Flat, or Down (downsloping).

- *Upsloping:* $m > 0.5 \frac{V}{s}$. (Often seen in normal, healthy exercise responses, even if slight ST depression is present).
- *Downsloping:* $m < -0.5 \frac{V}{s}$. (A highly specific indicator of severe ischemic heart disease).
- *Flat:* $|m| \leq 0.5 \frac{V}{s}$. (Also a strong indicator of ischemia, as a healthy ST segment should naturally slope upward into the T-wave).

3.2.3.4 Resting Electrocardiogram (RestingECG)

Definition: RestingECG is categorized into Normal, ST (having ST-T wave abnormality, such as T-wave inversion or ST elevation/depression > 0.05 mV), and LVH (showing probable or definite left ventricular hypertrophy by Estes' criteria).

T-wave Difficulties: T-waves are low-frequency, low-amplitude signals that are highly susceptible to baseline wandering, which is a type of low-frequency noise primarily caused by the user's respiration and minor electrode movements.

Solution:

1. Initial Attempt: To stabilize the T-wave for accurate inversion detection, we integrated a Wavelet Transform-based Baseline Wandering Removal algorithm (utilizing the open-source py-bwr library).

2. **The Trade-off:** While BWR successfully clarified the T-wave inversion morphology, testing revealed a critical flaw: the mathematical transformation aggressively warped the raw voltage levels, severely degrading the accuracy of ST elevation/depression measurements. Overall model accuracy dropped from 92.39% without BWR to 83.42% with global BWR.
3. **Final Pipeline:** To resolve this, the system employs a **bifurcated processing strategy**. Baseline correction is applied only in parallel streams specifically evaluating T-wave morphology, while **the original minimally filtered signal (0.5–35 Hz) is strictly preserved for calculating absolute ST segment offsets**.

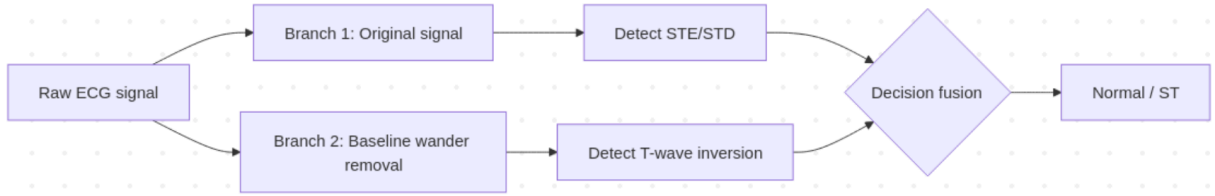


Figure 5: The bifurcated processing strategy for T-wave morphology correction.

LVH Difficulties: And for LVH, since it requires precordial leads—specifically V1, V5, and V6, we implemented a practical software-based workaround. During the initial profile registration on our web interface, the system proactively prompts users to input their known medical history, including any prior LVH diagnoses. This approach ensures that the machine learning model still receives the complete data array required for a comprehensive risk assessment.

3.2.3.5 Validation:

We validated the accuracy of the STE/STD feature by comparing it with the European ST-T Database. **The accuracy of recognizing ST Elevation/Depression was found to be 99.16% and recall of ST Elevation/Depression was 91.6%.**

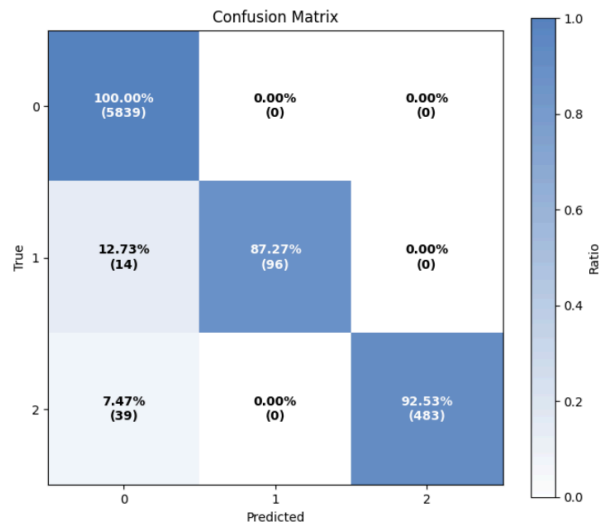


Figure 6: Confusion matrix for ST Elevation/Depression recognition.

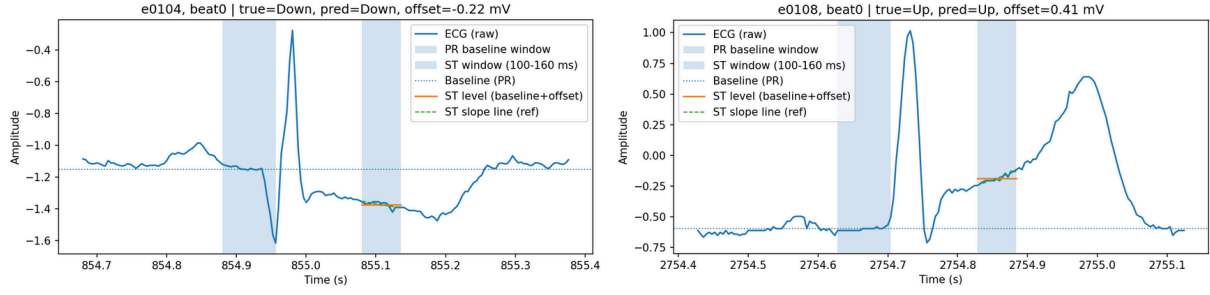


Figure 7: Plots of ECG signal with features annotated.

3.2.4 Model Training Methodology and Techniques

The final deployed 10-feature CatBoost model achieved **96.57% accuracy and 97.06% recall** on the holdout test set after fine-tuning and threshold tuning, demonstrating that ECG-derived features can be translated into reliable cardiovascular risk predictions.

3.2.4.1 Feature Preprocessing

First, public heart disease datasets were reconstructed into a comprehensive labeled dataset (918 records) with standardized feature naming and data types. Categorical variables were transformed into numerical representations via label encoding, while missing numerical values were handled using mean imputation to prevent training bias resulting from data omissions. To address the issue of class imbalance, oversampling was employed during the training phase. This approach ensures a more balanced distribution of positive and negative classes, thereby enhancing the model's ability to identify minority classes.

3.2.4.2 Training

Regarding data partitioning, this study adopted an 80/20 holdout split (80% training set; 20% test set) consistent with existing literature, performing cross-validation and hyperparameter optimization within the training set. A progressive evolution strategy was applied to model development; initially, multiple baseline models were evaluated, including Logistic Regression, Gradient Boosting, CatBoost, and an average probability Ensemble model. Through comprehensive comparison, CatBoost exhibited the highest potential. Consequently, the final deployed model in this study evolved directly from the CatBoost baseline. By performing further fine-tuning, we effectively improved the consistency between offline evaluation and deployment behavior. The final deployed CatBoost model converged with the following hyperparameters: (iterations=600, learning_rate=0.1, depth=6, l2_leaf_reg=3, random_seed=42).

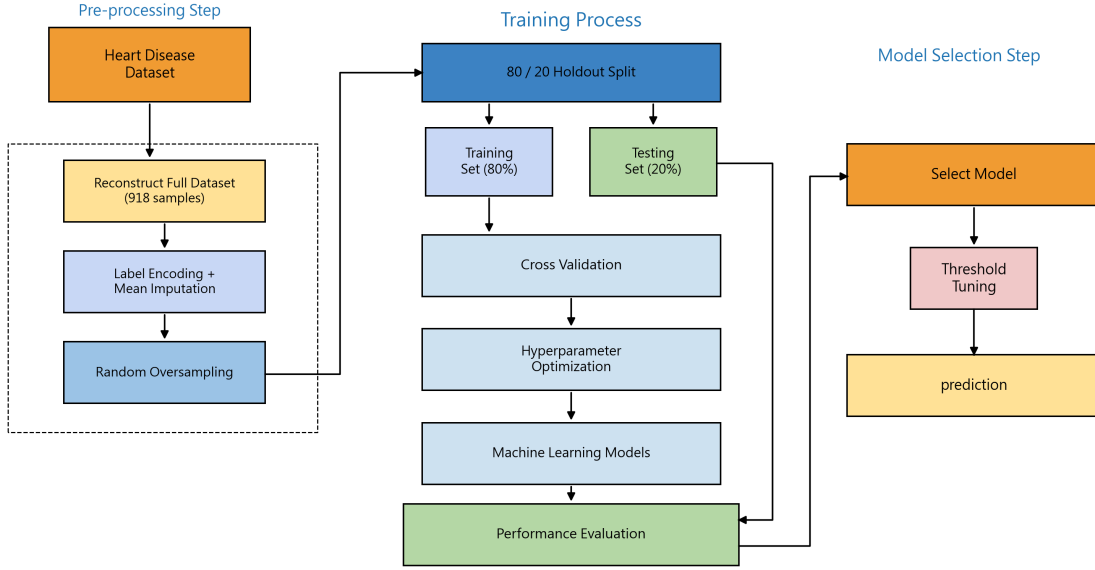


Figure 8: Flowchart of Model Development and Training

3.2.4.3 Classification

To balance prediction accuracy and system robustness, a **Dual-model Routing mechanism** was designed for the inference stage. When a user provides a complete set of clinical features, the system prioritizes the 10-Feature Model to deliver the best predictive performance. Conversely, if specific fields are missing, the system automatically switches to the 8-Feature Model, which utilizes data directly captured by our device's sensors to ensure stable and continuous output even under conditions of incomplete information.

3.3 AF Detection Methodology and Validation

During the early development of the AF detection module, the Coefficient of Variation (CV) test method was initially employed [7]. While this method is fast to implement and computationally efficient, its decision criteria are fundamentally based on a single statistic. This makes it prone to misidentifying ectopic rhythms, such as Premature Ventricular Contractions (PVCs), as AF, leading to a high false alarm rate in practical applications.

To address this issue, we subsequently explored a multi-feature integration approach proposed by Dash et al. [8], which utilizes TPR, RMSSD, and Shannon Entropy (SE) to improve discriminatory power. As shown in the comparative results, this integrated approach achieved excellent classification performance on the AFDB dataset. However, this method involves complex feature extraction and precise parameter tuning, incurring high engineering overhead and significantly longer computation times.

After a comprehensive evaluation of classification performance, false alarm control, algorithmic complexity, and maintenance costs, the RdR+NEC method [3] was ultimately selected for this work. This method offers the dual advantages of simple decision rules and **extremely low computational complexity**.

In the deployed system, **the final RdR+NEC version achieves 95.73% accuracy, 95.72% sensitivity, 95.74% specificity, and 1.167 ms/window** while satisfying the requirements for real-time execution on edge devices.

The specific technical details and parameter settings of the RdR+NEC algorithm are as follows:

3.3.1 RdR Map Construction

Unlike the traditional Lorenz plot which utilizes a single dimension, this method maps successive RR intervals (RR_i) and their differences ($dRR_i = RR_i - RR_{i-1}$) simultaneously onto a two-dimensional coordinate system. This design captures both heart rate and heart rate variability (HRV) information. During AF episodes, irregular rhythms cause data points to scatter randomly across a wide area of the map; conversely, normal or regular rhythms typically present as dense, localized clusters.

3.3.2 Gridding and Non-empty Cells (NEC)

The system divides the RdR map into a two-dimensional grid. For each input segment containing 128 cardiac cycles (128-beat window), the number of Non-empty Cells (NEC) is calculated. This count quantifies the degree of dispersion to classify the rhythm as AF or non-AF.

3.3.3 Threshold Determination

Based on the optimization analysis in [3], an NEC threshold of 65 was determined to provide optimal performance for a 128-beat window configuration.

3.3.4 Edge Deployment Advantages

Compared to methods requiring extensive feature extraction or complex floating-point operations, **RdR+NEC relies solely on a single integer count of NECs for classification**. Its low computational overhead enables high-precision monitoring while satisfying the real-time and long-term monitoring requirements of battery-powered wearable devices.

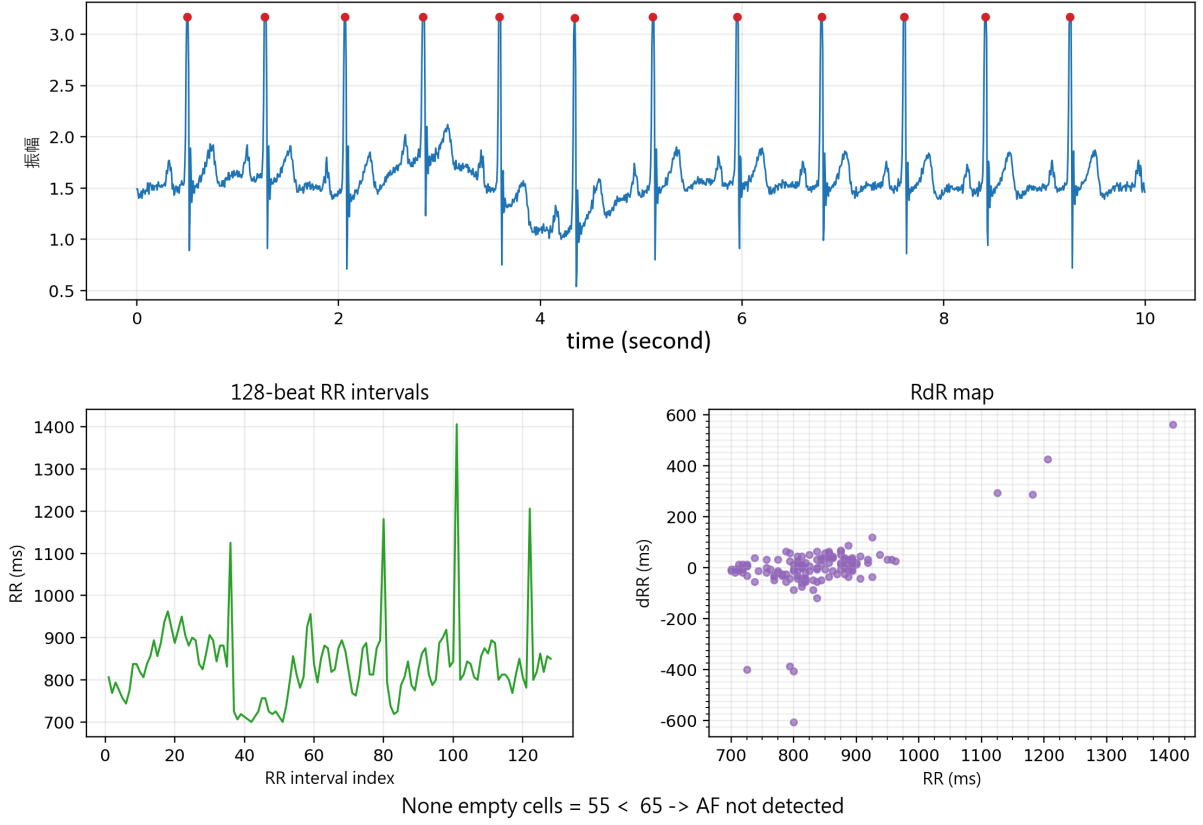


Figure 9: An example of Atrial Fibrillation (AF) Detection

To confirm the algorithm’s generalization capability, verification was conducted across multiple public datasets. The RdR+NEC method demonstrates stable and strong detection performance across diverse data distributions.

Performance validation results of the RdR+NEC method across public datasets

Table 3: Performance validation across public datasets

Database	Sensitivity	Specificity
NSRDB	NA*	88.10
NSR2DB	NA*	96.30
AFDB	95.70	95.70
Combined database	95.70	94.20

* NA: Sensitivity is not applicable because no true AF episodes are present.

3.4 TAIDE-based health assistant

To enhance user experience and provide personalized health insights, we integrated an advanced AI assistant directly into the frontend interface.

For the core language model, we selected **Gemma-3-TAIDE-12b-Chat**, a robust 12-billion parameter model optimized by the TAIDE project. This specific model was chosen for its excellent reasoning capabilities and its strong proficiency in handling Traditional Chinese,

ensuring that the health advice and explanations provided to users are both accurate and natural to read.

Due to the substantial computational requirements of running a 12B parameter model, we **deployed the language model as a dedicated server hosted on a Spark DGX system**. This backend server handles all the heavy AI inference tasks. When a user requests health insights on the frontend application, the system makes a call to the server, retrieving the AI-generated responses efficiently. **This client-server architecture ensures that the frontend remains highly responsive and accessible**, without being burdened by the intensive processing demands of the large language model.

```
system = (
    "你是健康助理。根據使用者的心臟健康指標，提供簡短、具體、非診斷性的建議。"
    "避免宣稱確診；若有危急徵象請提醒就醫。使用繁體中文。"
    "輸出請以 2~3 點條列為主，最後加一句提醒此為一般建議非醫療診斷。"
    "輸出格式為純文字，不使用任何styling。"
)
```

Snippet 1: TAIDE-based health assistant system prompt



Figure 10: TAIDE-based health assistant response

3.5 System Deployment, Cost Analysis, and Feasibility

To evaluate the practical feasibility of the proposed system, we analyze the hardware cost structure and deployment requirements.

The system consists of the following key components:

Table 4: Hardware Cost Breakdown of the Proposed System

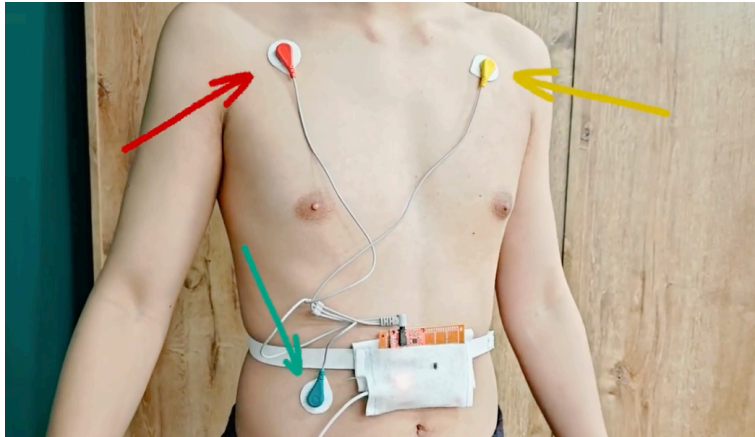
Component	Function	Cost (TWD)
AD8232 ECG module	ECG signal acquisition	~200
ESP32	Data acquisition and transmission	~150
Raspberry Pi 3B	Edge processing	~1,500
Power module & peripherals	Battery and supporting components	~100

Total		< 2,000
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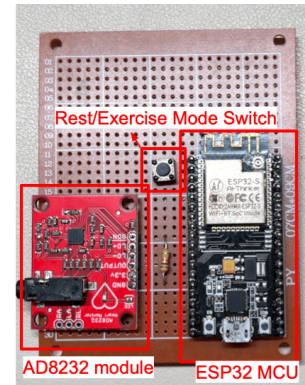
The total system cost is maintained below 2000 TWD, which is significantly lower than most commercial wearable ECG devices. From a system perspective, the proposed design achieves a favorable trade-off between computational capability, model complexity, and deployment cost. The use of lightweight machine learning models (approximately 650 KB) enables real-time inference on edge devices without requiring specialized hardware acceleration. Compared to smartwatch-based ECG systems, which typically rely on proprietary hardware ecosystems and external processing pipelines, the proposed system provides a transparent and cost-efficient architecture with full control over signal processing and inference. These results indicate that the proposed system is both technically effective and economically viable for large-scale deployment in home healthcare and remote monitoring scenarios.

3.6 Prototype Implementation and System Demonstration

The complete system consists of a wearable sensing unit and an edge-computing backend. On the hardware side, the chest-worn configuration combines AD8232 for single-lead ECG acquisition with ESP32 for sampling and wireless transmission (Figure 11a, Figure 11b). On the edge side, Raspberry Pi 3B executes signal preprocessing, AF screening, cardiovascular risk inference, and database-backed service orchestration for multi-user household use.



(a) Wearable ECG System Configuration and Electrode Placement



(b) Hardware Implementation of the Wearable ECG Monitoring System

Figure 11: Hardware Design and Wearable Configuration

In practical operation, the user first attaches the chest strap and ECG electrodes according to the predefined placement (Figure 11a). After device initialization, the measurement mode can be switched between resting and exercise conditions, enabling 24-hour ECG monitoring across daily activities rather than short static recordings. The frontend then provides real-time waveform visualization through WebSocket streaming (Figure 12), allowing immediate inspection of transient symptoms. This design supports on-demand recording of symptomatic episodes, and waveform plots can be exported for clinician review during outpatient visits.



Figure 12: Real-time ECG Waveform

The web interface is designed to support both initial onboarding and routine follow-up. At first login, users authenticate and complete personal profile and medical-history fields (Figure 15b); during subsequent sessions, daily measurement forms are submitted for longitudinal tracking (Figure 15a). The interface also provides light and dark theme modes for improved readability under different usage conditions.

For routine health interpretation, the frontend provides both trend charts and an integrated overview dashboard. Blood-pressure and heart-rate variation views are shown in Figure 13b and Figure 14a, while Figure 13a summarizes key indicators at a glance, including resting blood pressure, average heart rate, maximum heart rate, ST slope, RestingECG category, and AF-related rhythm status. Model-based cardiovascular risk scoring is presented in Figure 14b, where the quantified risk output is continuously synchronized with newly uploaded measurements.



(a) Overview



(b) Resting Blood Pressure

Figure 13: Frontend Interface



(a) Average Heart Rate



(b) Risk Evaluation

Figure 14: Frontend Interface

心悸寶貝
您的心臟健康小幫手
歡迎回來, 洪建偉 • 最後更新: 2026/4/1 下午6:25:51

總覽 靜息血壓 心率(1分鐘) 心率(30分鐘) 風險評估 即時ECG 健康數據

更新健康數據

靜息血壓 (mmHg)
例如: 120
正常範圍: 90-140 mmHg

血清膽固醇 (mm/dl)
例如: 200
正常範圍: 低於 200 mm/dl

空腹血糖 (mg/dl)
例如: 100
正常範圍: 70-100 mg/dl (空腹)

提交數據

數據說明

- 請定期更新您的健康數據, 以獲得更準確的健康評估。
- 建議每週至少更新一次數據。
- 數據將用於 AI 健康分析和風險評估。

AI 健康建議

根據您提供的心臟健康指標, 建議您:

- 留意運動時的心跳變化, 並觀察是否有異常。由於您的最大心率較高, 運動強度可能偏高, 請評估自身體力負荷。
- 注意血壓的變化, 維持規律作息與飲食均衡, 有助於血壓的穩定。
- ST段上升且向上傾斜的心電圖結果, 代表心臟的電氣活動有特定模式, 建議持續追蹤觀察。

提醒您, 以上為根據提供數據的一般性健康建議, 並非醫療診斷或治療方案。

(a) Daily Health Measurement

完善個人資料

請填寫以下資料以建立您的健康檔案

生理性別
請選擇

年齡
請輸入年齡

胸膈類別
請選擇

是否有運動心跳痛的問題
請選擇

是否曾經有被醫生診斷過左心室肥大 (LVH)
請選擇

提交資料

(b) First-time Profile Registration

Figure 15: Frontend Interface

To improve usability for non-specialist users, the system additionally provides AI-assisted health guidance in the interface. Based on the latest physiological indicators and risk outputs, the LLM module returns concise, non-diagnostic recommendations (Figure 10), helping users understand their condition and decide whether further medical consultation is needed.

Together, these results confirm that the proposed system is not only algorithmically effective but also practically deployable as a complete home-care monitoring workflow.

4 Research Outcomes and Benefits

4.1 Enhancement of Early Detection Capability of Cardiovascular Abnormalities

The proposed system significantly enhances early detection capability for cardiovascular abnormalities by enabling continuous, long-term ECG monitoring.

Conventional wearable ECG systems are typically limited to short-duration or user-triggered recordings, which may fail to capture transient or intermittent cardiac events, such as

paroxysmal AF. This limitation reduces the likelihood of early-stage detection, particularly for conditions that exhibit irregular temporal patterns.

In contrast, the proposed system supports continuous ECG acquisition and incorporates a rhythm analysis mechanism based on RR interval dynamics. By utilizing the RdR + NEC method, the system is able to effectively characterize irregular heartbeat patterns over extended time windows.

The ESP32 samples ECG continuously and transmits the data through a Wi-Fi link to the Raspberry Pi, where preprocessing and AF detection run in parallel with live visualization. The backend then streams ECG frames to the frontend through WebSocket, allowing users to observe real-time waveforms while the system analyzes rhythm irregularity.

The selected RdR+NEC demonstrates that the edge-based pipeline can balance early warning performance with low computational overhead. This design improves the likelihood of detecting subtle or sporadic abnormalities that would otherwise be missed in short-term measurements, and it provides a reliable foundation for preventive cardiovascular monitoring.

Table 5: AF detection method comparison on the MIT-BIH dataset

Method	Sensitivity	Specificity	Accuracy	Computation time
CV test	90.91	77.67	78.62	0.294 ms/window
RMSSD+TPR+SE	98.99	87.12	87.97	10.107 ms/window
RdR+NEC (selected method)	95.72	95.74	95.73	1.167 ms/window

4.2 From Raw ECG Signals to Interpretable Risk Representation

We introduce a unified feature extraction framework and a feature-based predictive model that map raw ECG signals to clinically relevant indicators for cardiovascular disease assessment. This representation enables the model to operate in a semantically meaningful feature space, rather than directly on high-dimensional raw signals. Furthermore, the feature-based approach reduces computational complexity, resulting in low overhead and enabling real-time execution on edge devices.

The final deployed 10-feature CatBoost model achieved 96.57% accuracy and 97.06% recall on the holdout test set after fine-tuning and threshold tuning. Precision, F1-score, and ROC-AUC also indicate that the extracted ECG features are sufficient for an interpretable and accurate risk model. Results on the holdout test set clearly illustrate the model’s evolution trajectory, starting from the initial CatBoost baseline model (accuracy 85.87%, recall 85.29%), moving through fine-tuning, and culminating in threshold tuning.

Table 6: Comparison of model performance metrics on the holdout test set

Model	Parameter	Accuracy	AUC	Precision	Recall	F1
CatBoost	{verbose=False, random_state=369}	85.87	90.35	88.78	85.29	87.00
LogisticRegression	{max_iter=5000, random_state=369}	87.50	92.53	89.11	88.24	88.67
GradientBoosting	{random_state=369}	85.33	91.69	90.32	82.35	86.15
Ensemble		86.96	92.06	90.62	85.29	87.88
CatBoost after fine-tuning (before threshold tuning)	(iterations=600, learning_rate=0.1, depth=6, l2_leaf_reg=3, random_seed=42)	91.67	88.08	95.70	87.25	91.28
CatBoost after fine-tuning (after threshold tuning)	(iterations=600, learning_rate=0.1, depth=6, l2_leaf_reg=3, random_seed=42, thresh- old=0.20)	96.57	97.67	96.12	97.06	96.59

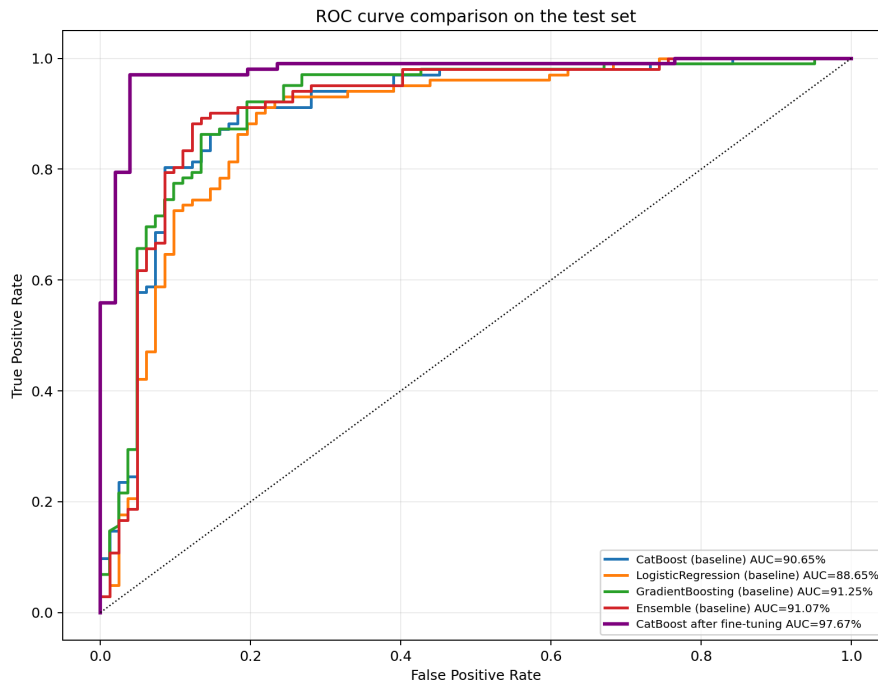


Figure 16: ROC curve comparison on test set

4.3 Lightweight Edge Backend Deployment

To evaluate deployment feasibility on resource-constrained hardware, we profiled the backend service on Raspberry Pi 3B under a representative user workflow. The device was first allowed to warm up at room temperature, and the Dockerized backend service was then launched at 19:30 after thermal stabilization. Five minutes later (19:35), interactive testing began with sequential operations including user login, health-data upload, chart inspection, live ECG viewing, and risk assessment.

As shown in Figure 17 and Figure 18, the backend maintained a consistently low computational footprint during routine operation. CPU utilization remained below 10% for most of the session, with only short transient increases during ECG chart rendering and risk-inference

execution. Thermal behavior was similarly stable: the Raspberry Pi temperature was maintained around 54°C throughout testing, corresponding to an increase of only approximately 4°C compared with the idle baseline.

These results indicate that the proposed backend service is sufficiently lightweight for sustained edge deployment. The system supports real-time interaction and model inference while preserving stable thermal and computational conditions, demonstrating practical suitability for long-term home and community healthcare scenarios.

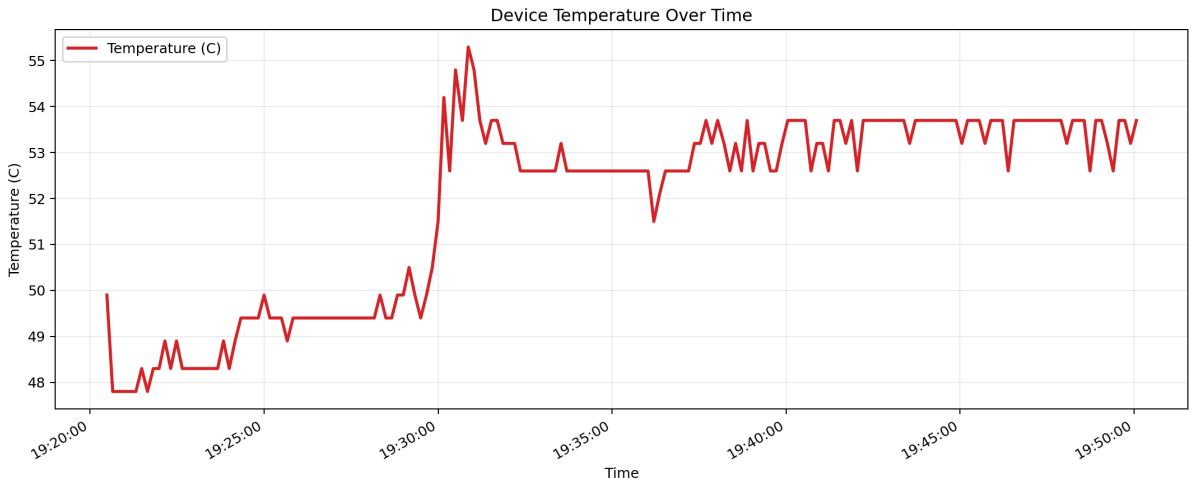


Figure 17: Raspberry Pi 3B's temperature over time

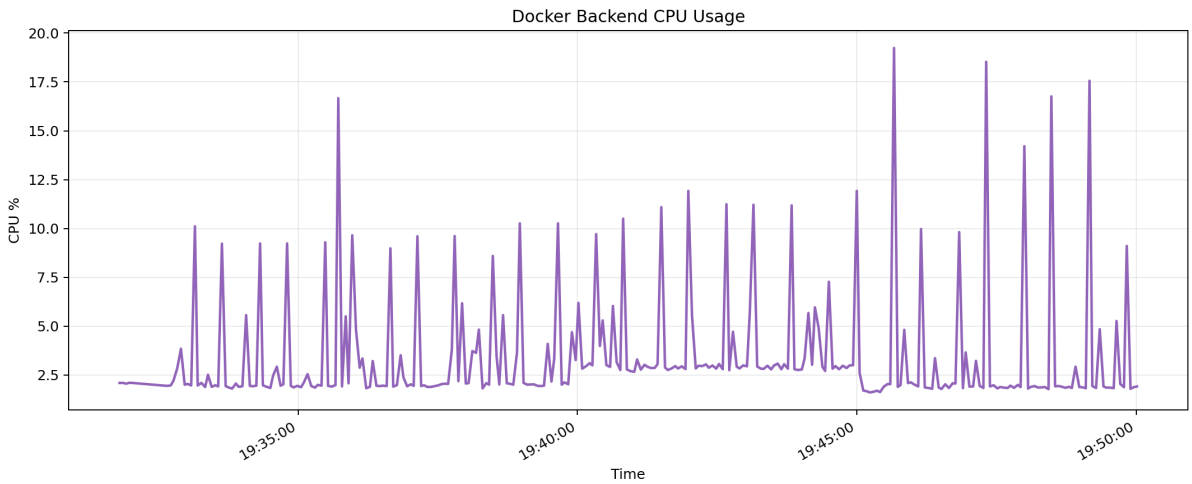


Figure 18: CPU usage of the service in Docker

5 Conclusion and Future Work

5.1 Conclusion

Our project presents a **highly integrated, low-cost, and edge-based wearable ECG monitoring system** designed to bridge the gap between expensive clinical diagnostics and

limited commercial smartwatches. By combining an AD8232 sensor, an ESP32 microcontroller, and a Raspberry Pi 3B, we established a robust hardware architecture **capable of long-term, continuous cardiovascular monitoring under both resting and exercise conditions.**

Algorithmically, the system successfully translates raw ECG signals into interpretable clinical indicators. The implementation of a **dual-model CatBoost routing mechanism ensures robust cardiovascular disease risk prediction (achieving up to 96.57% accuracy)** even with incomplete user data. Concurrently, the integration of the computationally efficient RdR+NEC algorithm **enables the real-time detection of both paroxysmal and persistent atrial fibrillation** with minimal computational overhead on edge devices. Complemented by a **TAIDE-based large language model**, the system empowers users with personalized and intuitive health insights. Overall, this framework offers an accessible, scalable, and privacy-preserving solution for proactive cardiovascular screening in home healthcare settings.

5.2 Future work

However, the proposed system still has several limitations. It currently supports only single-lead ECG, which may not be sufficient for all cardiovascular conditions. We plan to **expand to multi-lead ECG** in future versions to increase the information available for risk assessment. In addition, the current implementation still relies on gel-electrode contact for ECG measurement.

The system can be further extended to support multi-lead ECG acquisition, thereby enriching the information available for cardiovascular risk assessment. In addition, the current wired configuration and reliance on gel-based electrodes may limit user comfort and usability. Future work may therefore **investigate wireless electrode designs or dry electrode technologies** to enhance wearability and support long-term use. Furthermore, the system could be **integrated with cloud-based services to enable home-based diagnosis and remote monitoring**, allowing users to share ECG data and risk assessment results with healthcare professionals for more comprehensive and continuous care.

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